

- (c) Met $h = R/6$, toon aan dat $v_0 \sin \theta_0 = \sqrt{gR/3}$.

$$h = \frac{v_0^2 \sin^2 \theta_0}{2g}$$

With $h = R/6$, show that $v_0 \sin \theta_0 = \sqrt{gR/3}$.

$$v_0^2 \sin^2 \theta_0 = h \cdot 2g = (R/6) \cdot 2g = g(R/3)$$

$$v_0 \sin \theta_0 = \sqrt{gR/3}$$

(2)

- (d) Met die reikwydte $R = \frac{v_0^2 \sin 2\theta_0}{g}$ en die gegewens in (c), toon aan dat

$$v_0 \cos \theta_0 = \frac{1}{2} \sqrt{3gR} \text{ indien } \sin 2\theta = 2 \sin \theta \cos \theta.$$

With the range $R = \frac{v_0^2 \sin 2\theta_0}{g}$ and the formula given in (c), show that

$$v_0 \cos \theta_0 = \frac{1}{2} \sqrt{3gR} \text{ if } \sin 2\theta = 2 \sin \theta \cos \theta.$$

$$R = \frac{v_0^2 \sin 2\theta}{g}$$

$$v_0^2 \cdot 2 \sin \theta \cos \theta = Rg$$

$$v_0^2 \cos \theta = \frac{1}{2} \frac{Rg}{\sin \theta}$$

(3)

- (e) Toon aan dat die tyd wat nodig is om die piekhoogte te bereik gegee word deur $t_p = \sqrt{R/3g}$.

Show that the time it takes to reach the top is given by $t_p = \sqrt{R/3g}$.

$$T = \frac{2 v_0 \sin \theta_0}{g}$$

$$\frac{T}{2} = \frac{2 (\sqrt{gR/3})}{g}$$

$$t = \frac{2 (gR/3)^{1/2}}{g}$$

$$2 (g^{1/2})^{1/2} = t g^{1/2} \quad \frac{2 (gR/3)^{1/2}}{g} = t$$

$$\frac{2 \sqrt{R}}{\sqrt{3g}} = t$$

(3)